Martand Prakhar

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Summary

Information Science undergraduate (Class of 2028) with end to end full stack capabilities specializing in backend architecture, scalable REST APIs and database engineering. Hands on experience containerizing distributed multi service workflows with Docker and building real time vision pipelines.

Technical Skills

Languages: C++, C, TypeScript, JavaScript (ES6+), Python, SQL, HTML5, CSS3
Frontend: React.js, Vite, Tailwind CSS, Framer Motion, Recharts, Material UI
Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, WebSockets, JWT, Passport.js, Joi
Databases & DevOps: PostgreSQL, PostGIS, MongoDB, Mongoose, Prisma, Docker, Docker Compose, Nginx, Git
AI / CV & Tools: YOLOv8, OpenCV, PaddleOCR

Projects

AankhoDekha | **Demo Video** | **Live Site** | **GitHub** June 2026 – Present

- Built a multi camera ANPR platform transforming asynchronous camera feeds into vehicle trajectory tracking and deployed on AWS EC2.
- Integrated YOLOv8 plate detection with PaddleOCR and temporal clustering to eliminate duplicate reads with state code regex validation.
- Leveraged PostgreSQL with PostGIS for spatial trajectory reconstruction and cloned plate anomaly detection with streamed alerts via WebSockets.
- Accelerated video ingestion and inference throughput using device aware CUDA execution with YOLOv8 and PaddleOCR, applying temporal clustering to eliminate multi frame duplicate reads.
- Containerized services using Docker Compose and resolved memory bloat via multi stage builds.

Engineering Portfolio | **Live Site** | **GitHub** Oct 2026 - Present

- Built and deployed a React/TypeScript portfolio with project telemetry, DSA progress, and engineering incident documentation.
- Automated repository data synchronization using scheduled GitHub Actions workflows and scripts that generate portfolio JSON data.
- Integrated Recharts to visualize repository telemetry and DSA progress from generated JSON datasets.

MoneyTrack | **Live Site** | **GitHub** Oct 2025 – Present

- Developed a full stack personal finance tracker using unified transaction schemas, MongoDB models and secure JWT authenticated REST APIs.
- Implemented interactive cashflow visualizations with Recharts and automated tabular financial report exports to Excel.
- Containerized multi service frontend, backend, and database environments using Docker Compose with multi stage builds and Nginx serving.

Education

Dayananda Sagar College of Engineering Bengaluru, India
B.Tech in Information Science and Engineering | ***CGPA: 8.97 / 10*** *2024 – 2028*

- **Coursework:** Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks and Quant Aptitude

Algorithmic Core

Solved 350+ problems in C++ on LeetCode: Documenting my DSA journey on GitHub repository at: DSA_JOURNEY.